

The Technology Value Stream

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Course: CSD380 DevOPs, Module 1.2 Assignment

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Date: August 16, 2026

Introduction: What Is the Technology Value Stream?

- The Technology Value Stream represents the **end-to-end flow of work** from idea → development → deployment → customer value.
- It includes all activities required to deliver software: planning, coding, building, testing, deploying, and operating.
- DevOps focuses on **optimizing flow**, reducing delays, and improving reliability across this stream.
- Understanding the value stream helps organizations identify bottlenecks and accelerate delivery.

Why the Technology Value Stream Matters

- Aligns business goals with technical execution.
- Reveals inefficiencies such as long queues, manual approvals, and environment inconsistencies.
- Enables continuous improvement through automation and feedback loops.
- Supports rapid innovation, higher quality, and reduced operational risk.
- Forms the foundation for DevOps, Lean, and Agile transformation.

Lead Time vs. Processing Time

Lead Time

- Total elapsed time from request → delivery
- Includes waiting, approvals, handoffs, delays, and queues
- Reflects the **customer's experience**

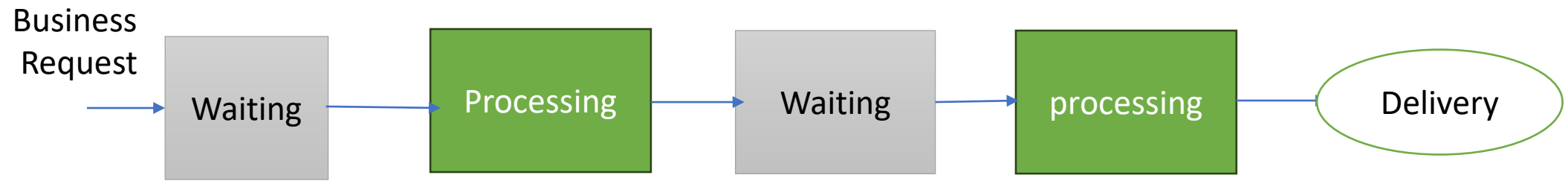
Processing Time

- Actual time spent actively working on the task
- Usually much shorter than lead time
- Reflects **engineering effort**

Key Insight

- Lead time is often **weeks or months**, while processing time is **hours**. This gap highlights where improvement is needed.

Diagram: Lead Time vs. Processing Time



Lead Time = Entire timeline (waiting + processing)

Processing Time = Only the active work segments

Common Scenario: Deployment Lead Times of Months

- Traditional organizations often experience deployment lead times of **weeks or months** due to:
- Manual testing and manual deployments
- Long approval chains (CAB meetings)
- Environment inconsistencies
- Siloed teams (Dev, QA, Ops, Security)
- Fear of breaking production → “big-bang releases”
- Limited automation and slow feedback loops

Consequences

- Slow delivery of customer value
- High deployment risk
- Increased failure rates
- Long recovery times
- Reduced innovation and competitiveness

DevOps Ideal: Deployment Lead Times of Minutes

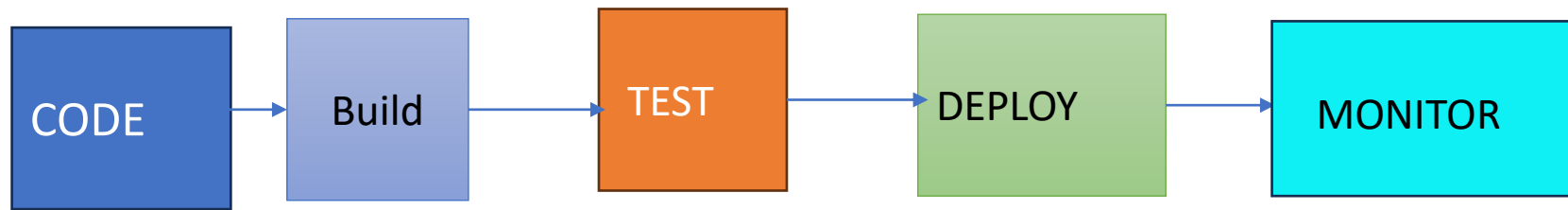
DevOps transforms deployment lead time from months → minutes using:

- Continuous Integration (CI) Automated builds and tests on every commit.
- Continuous Delivery/Deployment (CD) Automated pipelines that deploy safely and consistently.
- Infrastructure as Code (IaC) Reproducible environments using Terraform, Ansible, CloudFormation.
- Automated Testing Unit, integration, security, and performance tests.
- Monitoring & Feedback Loops Real-time insights into system health and user behavior.

Outcomes

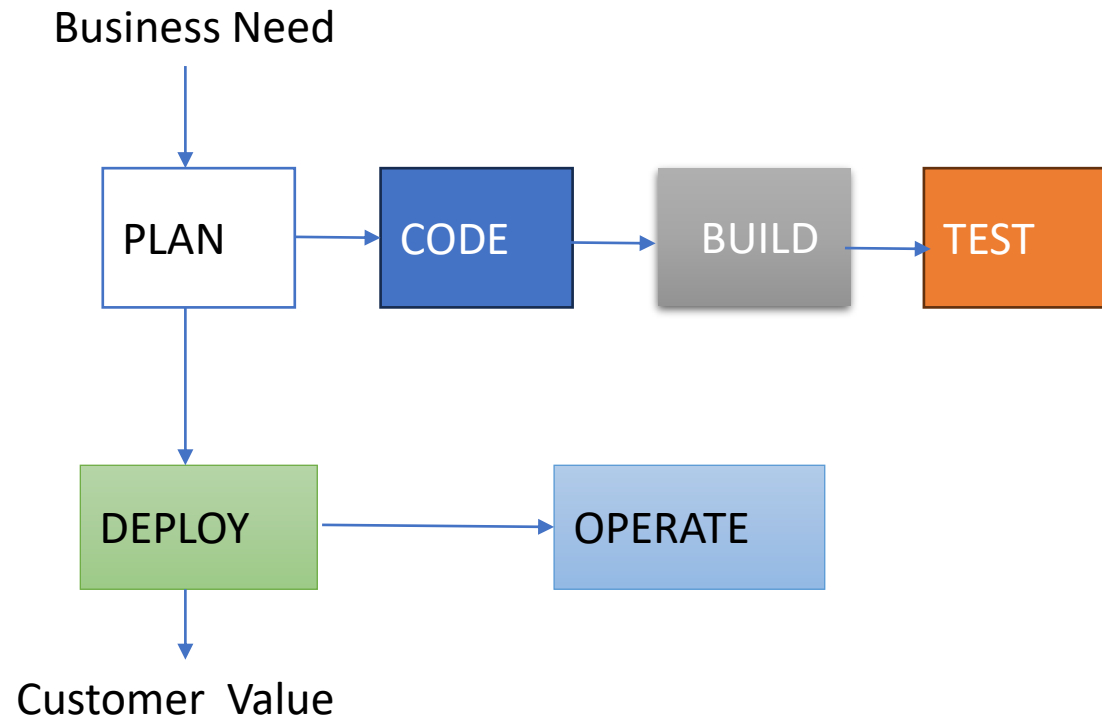
- Rapid innovation
- Safer releases
- Faster recovery (MTTR)
- Higher customer satisfaction

Diagram: DevOps Deployment Pipeline



Deployment Lead Time: ****Minutes****

Technology Value Stream Diagram (Full Flow)



Conclusion

- The Technology Value Stream reveals where delays occur.
- Lead Time vs. Processing Time highlights inefficiencies.
- Traditional deployment cycles are slow and risky.
- DevOps transforms delivery speed from **months** → **minutes**.
- Organizations gain agility, reliability, and customer satisfaction.
- Optimizing the value stream is essential for modern software delivery.

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